→ Interpolation

practice sheet 6

L8 : Chachy's Theorem

$f(x) = __ + __ + __ + __ + \dots$

$P_n(x) = __ + __ + __ + \dots$

→ Truncation error

Truncation error / Maximum error / Upper bound error

$$= \frac{f^{n+1}(\varepsilon)(x-x_0)(x-x_1)\dots(x-x_n)}{(n+1)!}$$

$\varepsilon \in [-1, 1]$

$x \in [\frac{\pi}{2}, 0, \pi]$

$f(x) = \cos x$

Find upper bound error.

Ans:

x	$f(x) = \cos x$
$x_0 = \pi/2$	0
$x_1 = 0$	1
$x_2 = \pi$	-1

So, $n = 2$

$$\text{Upper bound error} = \frac{f'''(\varepsilon)(x-x_0)(x-x_1)(x-x_2)}{3!}$$

$f(x) = \cos x$

$f'(x) = -\sin x$

$f''(x) = -\cos x$

$f'''(x) = \sin x$

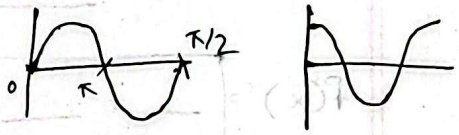
$$= \left|\frac{\sin\varepsilon}{3!}\right| \times \left|(x-\tfrac{\pi}{2})(x-0)(x-\pi)\right|$$

Max error এর জন্য $a \times b \rightarrow$ দুইটাই max

ε part

$w(x)$ part

ξ part's maximum value:



$$\xi \in [-1, 1]$$

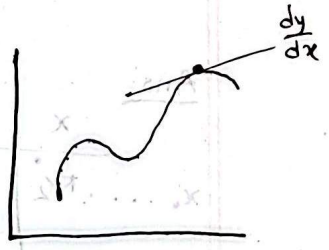
For -1, $\sin(-1) = -0.842$

For 0, $\sin(0) = 0$

For 1, $\sin(1) = 0.842$ [Max]

$$\left| \frac{\sin(1)}{3!} \right| = \left| \frac{0.842}{3!} \right| = 0.14$$

$\omega(x)$ part:



Max point of $\frac{dy}{dx} = 0$

$$\omega(x) = \left(x - \frac{\pi}{2}\right) x \left(x - \pi\right)$$

$$= x\left(x^2 - \frac{\pi}{2}x - \pi x + \frac{\pi^2}{2}\right)$$

$$= x^3 - \frac{\pi}{2}x^2 - \pi x^2 + \frac{\pi^2}{2}x$$

$$\omega'(x) = 3x^2 - 3\pi x + \frac{\pi^2}{2}$$

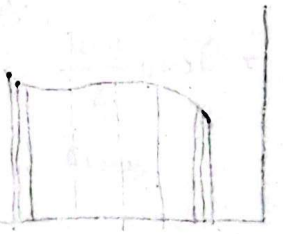
$$= \left| (x-\pi)(0-x)\left(\frac{\pi}{2}-x\right) \right| \times \left| \frac{3\sin\theta}{1\theta} \right| =$$

$$w'(x) = 0$$

$$\Rightarrow 3x^2 - 3\pi x + \frac{\pi}{2} = 0$$

$$\Rightarrow x = \frac{\pi}{2}\left(1 \pm \frac{1}{\sqrt{3}}\right)$$

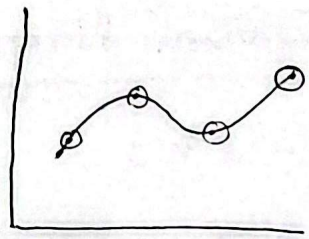
	x	$\omega(x) = \left\|(x-\frac{\pi}{2})x(x-\pi)\right\|$
$\omega(x)$	$\frac{\pi}{2}(1+\frac{1}{\sqrt{3}})$	1.4919
	$\frac{\pi}{2}(1-\frac{1}{\sqrt{3}})$	-1.4919
ε	-1	1.22
	1	10.64 [Max]

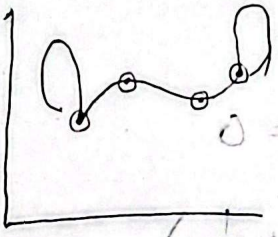


Upper bound error $= 0.14 \times 10.64$

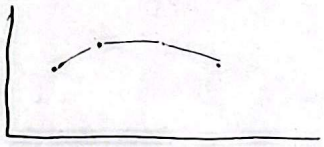
$= 1.489$

Why runge occurs? → Depends on the function (Runge/Symmetric function)
→ Depends on the nodes





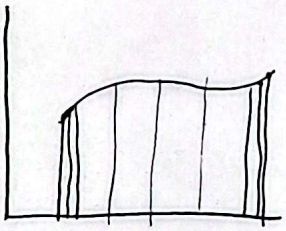
Runge Phenomenon
[২ পাশে ২টা edge create করে]



* Edge গুলোতে node বাড়িয়ে দিবে → Piece wise interpolation
solve হবে অনেকটা [Solution 1]

* Equal space দিয়ে nodes নিবো না [Solution 2]

Chebyshev Theorem

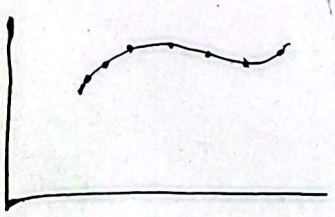


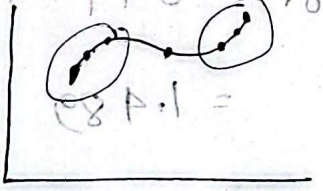
$A = \frac{b-a}{2}$ [Radius]

$\phi_j = \frac{(2j+1)\pi}{2(n+1)}$ [Angled point]

$x_j = A \cos \frac{(2j+1)\pi}{2(n+1)}$

Limitation: Only applicable in Range Function





✓

$f(x) = \frac{1}{16x}$

$n = 2\ (x_0, x_1, x_2)$

interval $= [-1, 1]$

(a) Find equal angled points / ϕ_s

(b) Find chebyshev node

(c) Find l_0.

x	$f(x)$

দেওয়া নাই Chevchn's Theorem use করবা

(a) Equal angled ϕ

$$\phi_0 = \frac{(2\times0+1)\pi}{2(2+1)} = \left(\frac{\pi}{6}\right)$$

$$\phi_1 = \frac{(2\times1+1)\pi}{2(2+1)} = \left(\frac{3\pi}{6}\right)$$

$$\phi_2 = \frac{(2\times2+1)\pi}{2(2+1)} = \left(\frac{5\pi}{6}\right)$$

(b)

x	$f(x) = \frac{1}{16x}$
$1\cos\frac{\pi}{6}$	$\frac{1}{16(\cos\frac{\pi}{6})} = 0.07$
$1\cos\left(\frac{3\pi}{6}\right)$	$\frac{1}{16\left(\cos\left(\frac{3\pi}{6}\right)\right)} = \infty$
$1\cos\left(\frac{5\pi}{6}\right)$	$\frac{1}{16\left(\cos\left(\frac{5\pi}{6}\right)\right)} = -0.05$

$$\frac{b-a}{2}\cos\theta + \frac{a+b}{2}$$

Radius Center

x	$f(x)=\sin^2\frac{x}{2}$
$-\pi/3$	$1/4$
0	0
$\pi/3$	$1/4$